

Calculus Review I: Inequalities, Limits, Derivatives, and More

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§1.1 Cauchy–Schwarz and Minkowski

The opening page recalls two inequalities:

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

and

$$\left(\sum_{i=1}^n (a_i + b_i)^2\right)^{1/2} \leq \left(\sum_{i=1}^n a_i^2\right)^{1/2} + \left(\sum_{i=1}^n b_i^2\right)^{1/2}.$$

The first is the Cauchy–Schwarz inequality; the second is Minkowski’s inequality, the triangle inequality in Euclidean norm.

The review function of these formulas is clear: they are often used to control error terms when proving limits.

§1.2 Neighborhood notation

For $x_0 \in \mathbb{R}$ and $\delta > 0$, the δ -neighborhood is

$$N_\delta(x_0) = \{x : |x - x_0| < \delta\}.$$

The punctured neighborhood is

$$\mathring{N}_\delta(x_0) = \{x : 0 < |x - x_0| < \delta\}.$$

The right and left neighborhoods are

$$N_\delta^+(x_0) = (x_0, x_0 + \delta), \quad N_\delta^-(x_0) = (x_0 - \delta, x_0).$$

For large variables, the notes use

$$U(\infty) = \{x : |x| > M\}, \quad U(+\infty) = \{x : x > M\}, \quad U(-\infty) = \{x : x < -M\}.$$

§1.3 Trigonometric identities

The first page lists a full table of trigonometric identities. The basic ones are

$$\sin^2 \alpha + \cos^2 \alpha = 1, \quad 1 + \tan^2 \alpha = \sec^2 \alpha, \quad 1 + \cot^2 \alpha = \csc^2 \alpha.$$

The addition formulas are

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta,$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta,$$

$$\operatorname{an}(\alpha \pm \beta) = \frac{\operatorname{an}\alpha \pm \operatorname{an}\beta}{1 \mp \operatorname{an}\alpha \operatorname{an}\beta}.$$

The sum-to-product and product-to-sum formulas appear as well, for example

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2},$$

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)].$$

The page includes double-angle and power-reduction formulas:

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha, \quad \cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha,$$

$$\sin^2 \alpha = \frac{1}{2}(1 - \cos 2\alpha), \quad \cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha).$$

§1.4 Hyperbolic functions

The next page introduces functions generated by e^x :

$$\sinh x = \frac{1}{2}(e^x - e^{-x}), \quad \cosh x = \frac{1}{2}(e^x + e^{-x}),$$

$$\operatorname{anh} x = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad \coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}.$$

Their identities mirror trigonometric identities but with sign changes:

$$\cosh^2 x - \sinh^2 x = 1,$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y,$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

The note treats them as a useful parallel system rather than as isolated special functions.

§1.5 Sequence limits: epsilon–N language

A sequence $\{x_n\}$ converges to A if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - A| < \varepsilon.$$

We write

$$\lim_{n \rightarrow \infty} x_n = A.$$

The note highlights the uniqueness theorem. If a sequence had two limits A and B , choose

$$\varepsilon = \frac{|A - B|}{2}.$$

For large n ,

$$|A - B| \leq |A - x_n| + |x_n - B| < 2\varepsilon = |A - B|,$$

a contradiction.

§1.6 Infinite limits and one-sided limits

A sequence tends to $+\infty$ if for every $G > 0$ there is N such that

$$n > N \implies x_n > G.$$

For functions, the epsilon-delta definition is

$$\lim_{x \rightarrow x_0} f(x) = A$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - x_0| < \delta \implies |f(x) - A| < \varepsilon.$$

One-sided limits are defined by replacing the punctured neighborhood with a right or left punctured neighborhood. A two-sided limit exists iff the two one-sided limits exist and are equal.

§1.7 Heine criterion for function limits

The highlighted theorem says:

$$\lim_{x \rightarrow x_0} f(x) = A$$

if and only if for every sequence $x_n \rightarrow x_0$ with $x_n \neq x_0$, one has

$$f(x_n) \rightarrow A.$$

This is often used to prove that a limit does not exist. For example,

$$\lim_{x \rightarrow 0} \cos \frac{1}{x}$$

does not exist because along suitable sequences the values tend to different limits. The scanned page uses the sequences

$$x_n = \frac{1}{2n\pi}, \quad y_n = \frac{1}{2n\pi + \pi/2}.$$

Then $x_n, y_n \rightarrow 0$, but $f(x_n) \rightarrow 1$ and $f(y_n) \rightarrow 0$.

§1.8 Infinitesimals and infinitesimal orders

An infinitesimal is a variable whose limit is zero. The note warns that a fixed small number such as e^{-100} is not an infinitesimal; infinitesimal means a variable in a limiting process.

The pages record the usual rules:

- a finite sum of infinitesimals is infinitesimal;
- a bounded variable times an infinitesimal is infinitesimal;
- a constant times an infinitesimal is infinitesimal;
- the product of two infinitesimals is infinitesimal;
- the product of infinitely many infinitesimals need not be handled by a naive rule.

The handwritten orange comment gives examples such as

$$1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots$$

and warns that infinitely many factors require separate analysis. The problem is not philosophical; it is about whether one is taking a finite algebraic combination or a changing infinite product.

For two infinitesimals α, β , one writes

$$\alpha = o(\beta) \iff \lim \frac{\alpha}{\beta} = 0,$$

and

$$\alpha \sim \beta \iff \lim \frac{\alpha}{\beta} = 1.$$

Equivalent infinitesimals may be substituted in products and quotients when the relevant limits exist.

§1.9 Squeeze theorem and monotone convergence

The squeeze theorem is recorded as:

$$X \leq Y \leq Z, \quad \lim X = \lim Z = A \implies \lim Y = A.$$

The monotone convergence theorem says that a monotone bounded sequence converges. For example, an increasing sequence with an upper bound converges to its least upper bound; a decreasing sequence with a lower bound converges to its greatest lower bound.

The page gives the standard proof for the decreasing case: let A be the infimum. Since A is the greatest lower bound, for every $\varepsilon > 0$ there exists some $x_N < A + \varepsilon$, and monotonicity then gives $A \leq x_n \leq x_N < A + \varepsilon$ for $n > N$.

§1.10 Cauchy criteria

The sequence Cauchy criterion is

$$\{x_n\} \text{extconverges} \iff \forall \varepsilon > 0, \exists N, m, n > N \Rightarrow |x_m - x_n| < \varepsilon.$$

For function limits, the analogous criterion is: $f(x)$ has a finite limit at x_0 iff for every $\varepsilon > 0$ there is a punctured neighborhood such that

$$x_1, x_2 \in \mathring{N}_\delta(x_0) \implies |f(x_1) - f(x_2)| < \varepsilon.$$

The note states that the proof of sufficiency uses completeness of the real numbers, so it is often skipped in elementary courses.

§1.11 Continuity

A function is continuous at x_0 if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

Equivalent sequentially:

$$x_n \rightarrow x_0 \implies f(x_n) \rightarrow f(x_0).$$

The pages list the usual operations: sums, products, quotients with nonzero denominator, and compositions of continuous functions are continuous where defined.

The note uses continuity to justify passing limits through elementary functions, for example

$$\lim \sin u_n = \sin(\lim u_n)$$

when u_n has a limit.

§1.12 Derivatives and differentials

The later pages review derivative definitions:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

The differential form is

$$dy = f'(x) dx.$$

The derivative rules are:

$$(u + v)' = u' + v', \quad (uv)' = u'v + uv', \quad (u/v)' = \frac{u'v - uv'}{v^2},$$

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

The scanned pages also include logarithmic differentiation and implicit differentiation examples. The practical method is to take logarithms when products, quotients, and powers are entangled, and to differentiate both sides when the function is defined implicitly.

§1.13 Taylor-type expansions

The final pages include expansion examples and reminders about higher derivatives. The standard Taylor form near $x = 0$ is

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots.$$

Frequently used expansions include

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots,$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

The handwritten derivations use these expansions to compare infinitesimals and compute limits.

§1.14 Where the calculation points

This note is a calculus foundation review. The chain of ideas is:

neighborhoods $\rightarrow$ epsilon definitions $\rightarrow$ sequence criteria
 $\rightarrow$ infinitesimals $\rightarrow$ continuity $\rightarrow$ derivatives and expansions.

The handwritten comments are mostly warnings: do not confuse small constants with infinitesimals; do not multiply infinitely many infinitesimals naively; do not prove nonexistence of a limit without constructing incompatible approaches; do not use equivalent infinitesimals outside their valid algebraic context.

§1.15 Limits through filters and sequences

The epsilon- N definition says that eventually all terms lie in every prescribed neighborhood. In metric spaces, sequences are enough to test limits. In general topological spaces, nets and filters replace sequences.

This explains the Heine criterion: for functions between metric spaces, a limit exists iff every convergent sequence approaching the point has the corresponding image limit.

§1.16 Hyperbolic functions and exponential coordinates

The hyperbolic functions

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

parametrize the hyperbola

$$\cosh^2 x - \sinh^2 x = 1.$$

They are to Lorentzian geometry what sine and cosine are to Euclidean circle geometry.

§1.17 Cauchy–Schwarz, Minkowski, and normed spaces

Cauchy–Schwarz gives inner-product control, while Minkowski gives the triangle inequality for norms. These inequalities are the finite-dimensional entrance to Banach and Hilbert spaces, where convergence, projection, and approximation become geometric.

§1.18 Calculus review through limits and norm geometry

The review topics are connected by structure: limits are neighborhood statements, trigonometric and hyperbolic identities come from group/exponential parametrizations, and inequalities express norm or inner-product geometry.